

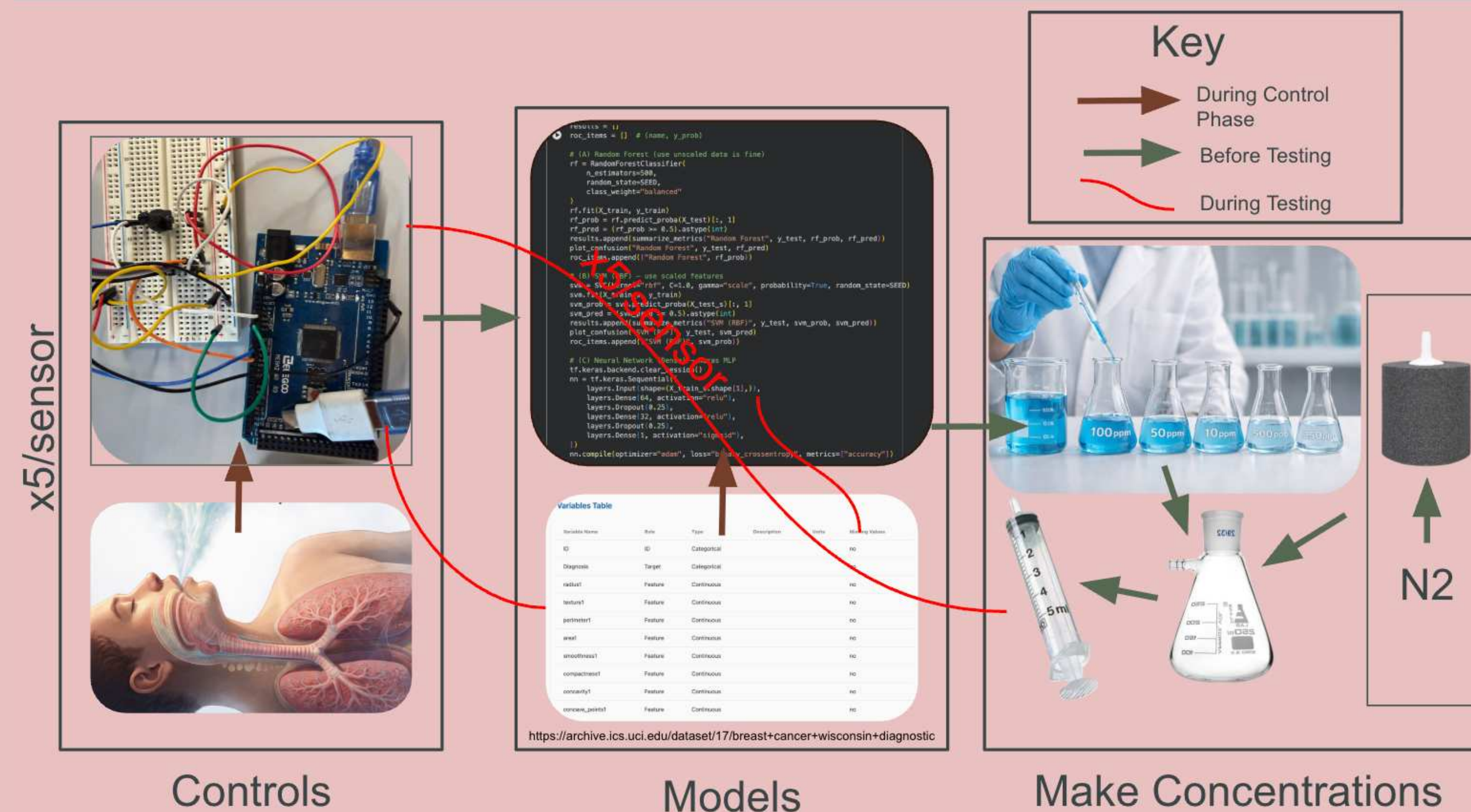
DETECTING LUNG CANCER VOCS USING A DIVERSE E-NOSE

RSEF Category: Material Science

Introduction

- **Lung cancer** is the leading cause of cancer-related deaths globally, causing 1.8 million deaths in 2020 (Shang et al., 2023).
- **Early detection is difficult** because initial symptoms like coughing and weight loss are often ignored or unnoticed.
- **Diagnostic methods** such as CT scans, MRIs, and biopsies are often costly and inaccessible, especially in developing countries.
- As a result, **85% of cases are diagnosed at a later stage**, when treatment is more difficult and less effective.
- **The current five-year survival rate is only 10–15%**, but early diagnosis could raise it to 80% (Shang et al., 2023).

Methodology



1. Build an **Arduino apparatus** compatible with MQ sensors
2. **Pulse room air** across sensors at set intervals
3. **Prepare ethanol dilutions** for testing
4. **Train an ML model** to predict ethanol concentration from sensor data
5. **Compare single sensor accuracy against the full array**

Conclusions

- These findings demonstrate that **each sensor differs significantly in its ethanol detection range ($p < 0.01$)**, and that **individual sensors classify best under an SVM (RBF) model** (average accuracy = 96%), with the **MQ2 and MQ6 sensors reaching 100% accuracy** on their own.
- **When data from all six sensors were combined**, both the **Random Forest and SVM (RBF) models achieved 100% accuracy** across all samples, and there was **no significant statistical difference between the full diverse E-Nose array and the strongest individual sensors**, supporting the conclusion that while a diverse E-Nose is **highly accurate**, certain **individual sensors perform comparably** which is not supported by the hypothesis.
- This **improvement is meaningful** as **sensors that flag diseases earlier** could help shift diagnoses into the curable window, where survival may climb toward 80%.

Purpose & Hypothesis

This study will investigate the effect of a Diverse E- Nose have for a specific lung cancer VOC (ethanol).

- A diverse sensor array has been shown in the past to help with the detection low concentration compounds (Saha et al., 2012)
- It is hypothesized that a machine learning model based on a multi-sensor array will outperform individual sensors across all concentrations
- IV: Sensor type & Ethanol concentrate
- DV: Model Accuracy
- Control: Individual Sensor voltage



Results

Image 1: MQ-2 voltage across time

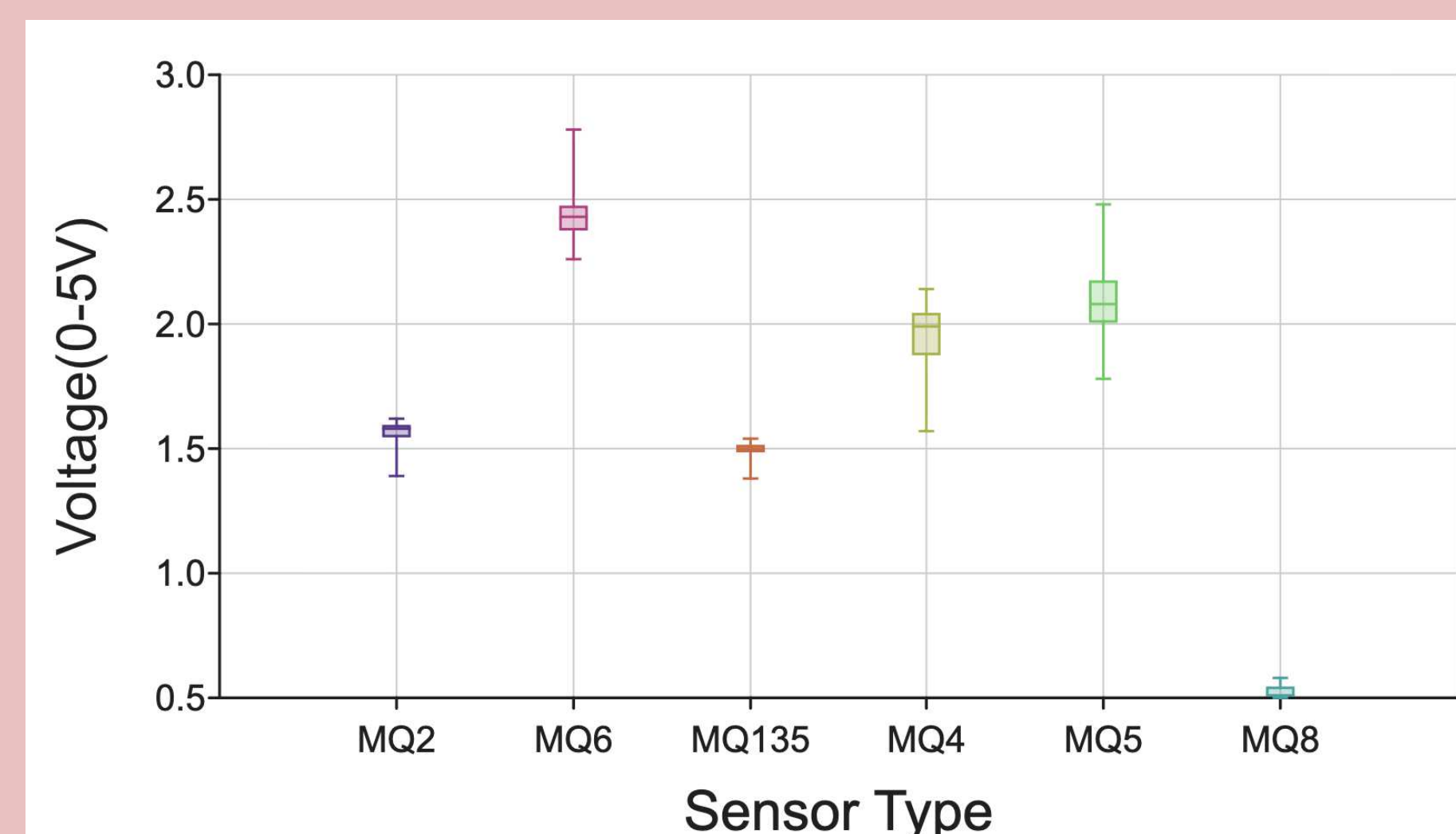


Image 2: Voltage Range for all Sensors

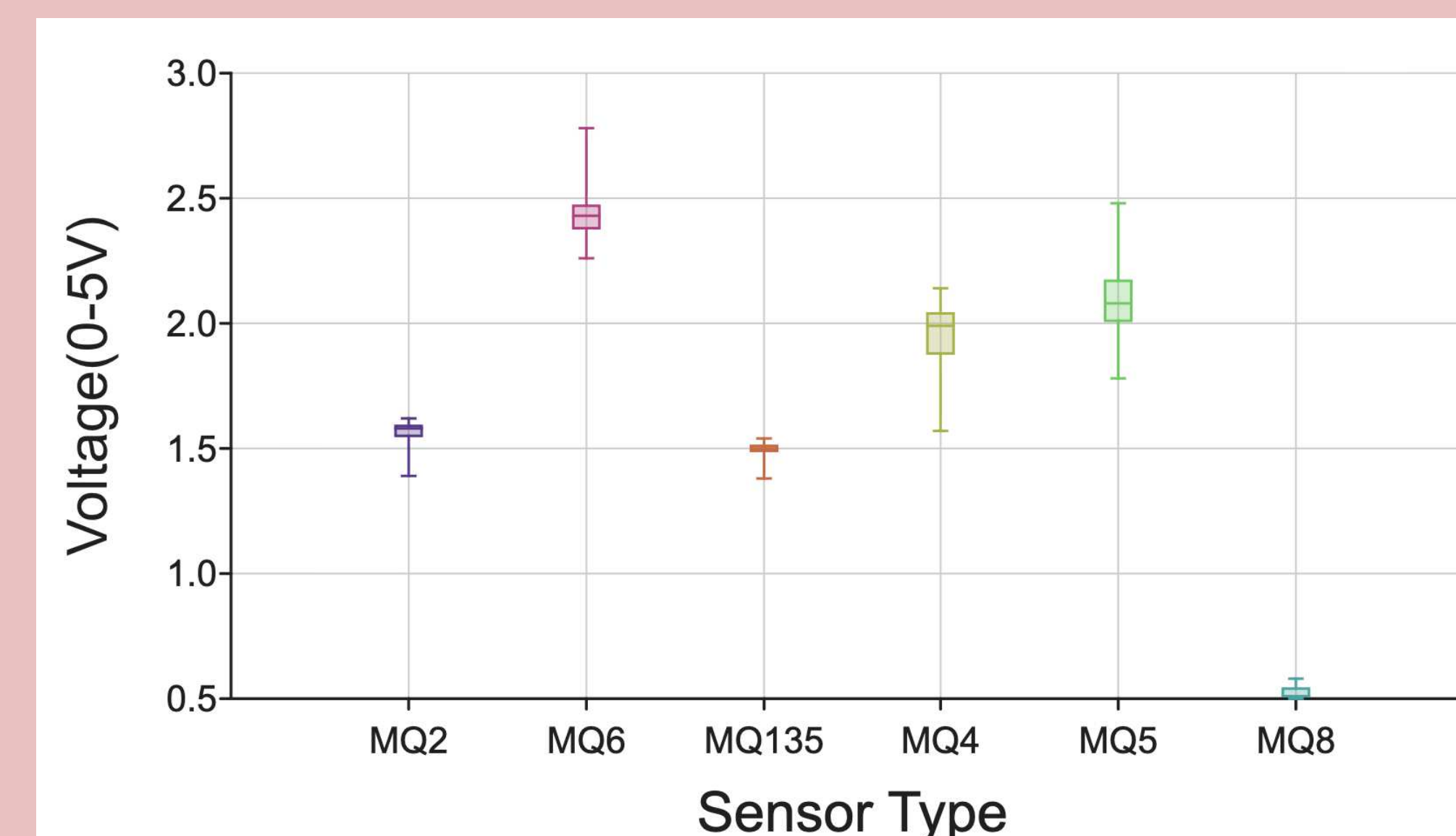


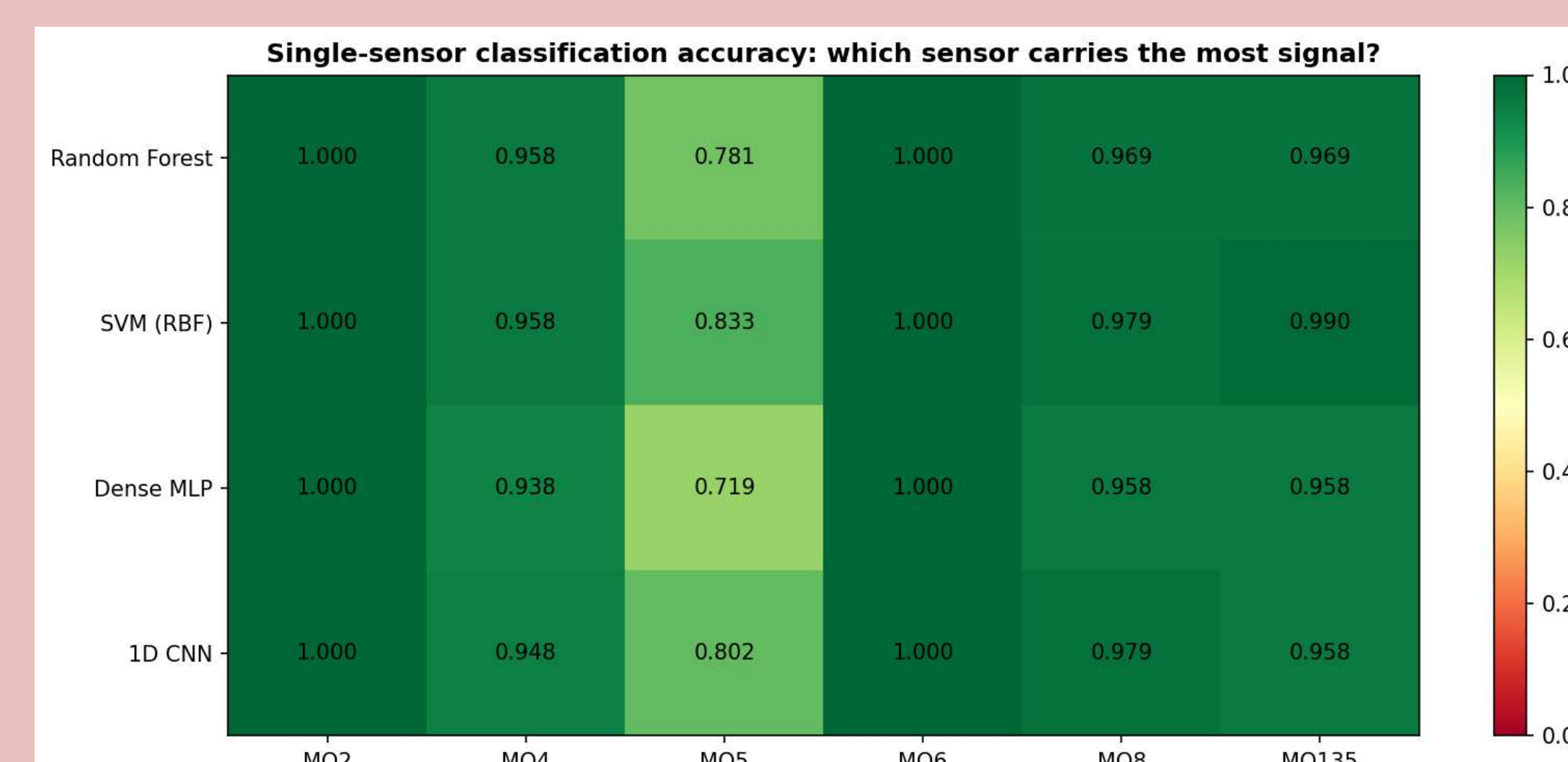
Image 3: ANOVA Test For Voltage Range

Effect	Degrees of Freedom (df)	F-statistic (MS / MS residual)	P-value
Sensor Type	5	21000	<0,01
Error or Residual	2079		

Image 5: ANOVA of classification accuracy, full array vs. top sensors (MQ2, MQ6)

One-way ANOVA per algorithm: Accuracy across {Full array, MQ2 only, MQ6 only}					
H0: the three sensor configurations produce equal mean accuracy					
Algorithm	Effect	df	F-statistic (MS / MS_residual)	P-value	Significant?
Random Forest	Sensor Configuration	2, 27	1.000	0.381	No
SVM (RBF)	Sensor Configuration	2, 27	1.260	0.3	No
Dense MLP	Sensor Configuration	2, 27	18.077	<0.001	Yes
1D CNN	Sensor Configuration	2, 12	0.991	0.4	No

Image 4: Model Accuracy Heatmap per sensors



References

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